Cybersecurity Practical – Questions & Commands

LJIET | Semester VII | All 23 practicals | Questions reproduced from the supplied PDF

1. Network Mapper (Nmap)

Task: Use Nmap to perform host discovery, port scanning, service and version detection, and OS detection on an authorized lab. Prepare a report including findings and defensive best practices.

```
nmap --version
nmap -sn 127.0.0.1
nmap 127.0.0.1
nmap -p 22,80,443 127.0.0.1
nmap -sV 127.0.0.1
sudo nmap -O 127.0.0.1
nmap -sV -oN nmap_report.txt 127.0.0.1
cat nmap_report.txt
```

2. Wireshark

Task: Capture, analyze, and interpret network packets for troubleshooting and security analysis. Apply an ICMP capture filter; examine DNS, HTTPS metadata, and the TCP three-way handshake. Use only designated training traffic.

```
sudo wireshark
# Capture filter: icmp
# Display filter: dns
# Display filter: tls
# Display filter: tcp.flags.syn == 1 || tcp.flags.ack == 1
```

3. John the Ripper (Password Hash Analysis)

Task: Analyze supplied sample hashes, identify likely hash algorithms, and discuss dictionary-attack effectiveness, impact, prevention, and mitigation. Use only provided lab hashes.

```
john --list=formats
john --format=RAW-MD5 --wordlist=/usr/share/wordlists/rockyou.txt sample_hashes.txt
john --show sample_hashes.txt
john --show --format=RAW-MD5 sample_hashes.txt
```

4. Whois (Domain Lookup)

Task: Retrieve publicly available domain registration details, registrar information, nameservers, privacy protection, and DNS information. Explain information-disclosure risks and legal/ethical limitations.

```
whois example.com
whois -h whois.iana.org example.com
dig example.com NS
```

5. Dig (DNS Query)

Task: Use dig to retrieve DNS records such as A, MX, and NS for an approved domain. Explain the returned information and the difference between active and passive information gathering.

```
dig example.com A
dig example.com MX
dig example.com NS
dig example.com AAAA
```

6. theHarvester (OSINT)

Task: Gather publicly available information such as email addresses and subdomains from approved public sources. Discuss security impact and ethics, and remain within the approved scope.

```
theHarvester -d example.com -b crtsh
theHarvester -d example.com -b duckduckgo
theHarvester -d example.com -b crtsh -f harvester_report.html
```

7. Zphisher (Phishing Awareness)

Task: In an isolated, authorized training environment, demonstrate how a fake login page can imitate a legitimate website. Use dummy data only and explain the technique, impact, and prevention.

```
# No credential-harvesting commands included.
# Use an instructor-approved phishing-awareness simulator with dummy data only.
```

8. Burp Suite Intruder

Task: Use an authorized training target and test data to understand a Sniper attack. Explain Sniper, Clusterbomb, Battering Ram, and Pitchfork, and document impact and defensive measures.

```
burpsuite
# GUI: Proxy > Intercept; send the authorized request to Intruder.
# GUI: Select Sniper, Clusterbomb, Battering Ram, or Pitchfork.
# GUI: Use instructor-provided test payloads and start the attack.
```

9. Sublist3r

Task: Enumerate subdomains of an authorized target. Include command syntax, save output to a file, and document findings and defensive measures.

```
sublist3r -d example.com
sublist3r -d example.com -o subdomains.txt
cat subdomains.txt
```

10. DirBuster (Directory/File Discovery)

Task: Discover directories and files on an authorized training website. Identify potentially sensitive paths, state the wordlist used, explain the relevant VAPT phase, and recommend prevention.

```
dirbuster
# GUI: Enter the authorized training URL.
# GUI: Select the instructor-approved wordlist.
# GUI: Start the scan and export the results.
```

11. Shodan (Internet-Exposed Asset Search)

Task: Search Shodan for exposed devices, ports, and publicly reported vulnerabilities. Explain potential risks and prepare a report with defensive recommendations.

```
# Shodan web interface: https://www.shodan.io/
# Search only assets you own or are authorized to assess.
# Example query for your own public IP: net:YOUR_PUBLIC_IP
# Review host details and export/save the authorized findings.
```

12. Hydra (Password-Attack Demonstration)

Task: In a controlled lab using the provided training machine, explain Hydra's purpose and syntax and perform a limited test only with approved test accounts. Include impact and mitigation.

```
# Use only the instructor-provided local lab and dummy account.
hydra -l testuser -P lab_passwords.txt 127.0.0.1 ssh -t 1 -V
# Stop the test after the approved limited attempt.
```

13. Dnsenum (DNS Information Gathering)

Task: Collect DNS information for an authorized target, including NS, MX, A, and AAAA records. Explain the information-gathering process and document findings.

```
dnsenum example.com
dnsenum --enum example.com
dnsenum --output dnsenum_report.xml example.com
```

14. Gobuster (Directory/File Discovery)

Task: Use Gobuster against an authorized training website to identify hidden files or directories. Explain its purpose, the relevant VAPT phase, and defensive recommendations.

```
gobuster dir -u http://127.0.0.1 -w /usr/share/wordlists/dirb/common.txt
gobuster dir -u http://127.0.0.1 -w /usr/share/wordlists/dirb/common.txt -o gobuster.txt
cat gobuster.txt
```

15. Burp Suite (DVWA Login Testing)

Task: Configure the Burp Suite proxy and inspect a controlled authentication request on the local DVWA login page using dummy credentials. Report evidence, impact, and mitigation.

```
burpsuite
# Open DVWA locally in the browser and configure its proxy to 127.0.0.1:8080.
# Submit dummy credentials and inspect the request in Proxy > HTTP history.
# Save the authorized request/response evidence.
```

16. Linux File and Directory Management Commands

Task: Practice pwd, ls, cd, mkdir, cp, mv, rm, and find. Explain each command, run examples in a practice directory, and record the output.

```
pwd
ls
mkdir practice
cd practice
cp /etc/hostname hostname_copy.txt
mv hostname_copy.txt renamed.txt
find . -type f
rm renamed.txt
```

17. Linux File Permissions and User Management

Task: Practice chmod, chown, whoami, id, groups, and ls -l. Examine file permissions and explain least privilege and safe permission settings.

```
whoami
id
groups
touch permission_test.txt
ls -l permission_test.txt
chmod 600 permission_test.txt
sudo chown "$USER":"$(id -gn)" permission_test.txt
```

18. Command Injection on DVWA

Task: In the local DVWA lab, demonstrate command injection using a harmless test command. Record the result and explain prevention, mitigation, impact, and best practices.

```
# In DVWA Security set the level as instructed by your lab supervisor.
# In the DVWA Command Injection field, use this harmless local test:
127.0.0.1; whoami
# Record the displayed result in the local lab only.
```

19. Linux Process and System Monitoring Commands

Task: Use ps, top, free, df, du, uptime, and a safe test process to inspect running processes and system resources. Record the commands and explain their output.

```
ps
top
free -h
df -h
du -sh .
uptime
sleep 300 &
```

20. Windows Command-Line Networking and Diagnostic Commands

Task: Practice ipconfig, ping, tracert, nslookup, netstat, and arp. Record command outputs and explain how each supports network troubleshooting.

```
ipconfig
ping 127.0.0.1
tracert 127.0.0.1
nslookup example.com
netstat -ano
arp -a
```

21. SQL Injection on DVWA and Hash Analysis

Task: Set up the local training application and test SQL injection using lab data only. Discuss secure query practices and analyze sample lab hashes where appropriate.

```
# Open the local DVWA SQL Injection module using dummy lab data.
# Basic training input (low-security DVWA lab only):
1' OR '1'='1' #
# Analyze only the supplied sample hash file:
john --show sample_hashes.txt
```

22. OpenVAS / Greenbone Target and Task Creation

Task: Create a target for an authorized lab machine, configure a scan task, select an available scan configuration, start the task, and record its settings and status.

```
sudo gvm-setup
sudo gvm-check-setup
sudo gvm-start
# Open the Greenbone web interface and create the authorized target and scan task.
```

23. OpenVAS Scan Results Analysis and Report Generation

Task: Review scan findings, severity, affected hosts and ports, CVE references where available, and remediation guidance. Verify findings where needed and export a report.

```
sudo gvm-check-setup
sudo gvm-start
# In Greenbone: open the completed task > Reports.
# Review findings and export the report in the required format.
```